

A TRANSLATED-CUSP BOUNDARY SET OF POSITIVE HARMONIC MEASURE

ARGUS OPEN RESEARCH PROJECT

ABSTRACT. Problem 7.15 of the Canterbury problem collection asks for which $\alpha \in (1/2, 1)$ a boundary set defined by local containment in translates of

$$C_\alpha = \{x + iy : 0 < x < |y|^\alpha\}$$

must have harmonic measure zero. The printed replacement clause does not separately specify how the distinguished vertex of the preceding angular definition is transported. We therefore state the conclusion relative to an explicit source-vocabulary convention: the single comparison object may be replaced, retained, weakened to boundary contact, or expanded at the same parameter, but no new atomic predicate and no second-parameter copy may be introduced. Under every endpoint-consistent normal form in this convention, we give a bounded Jordan domain for which the resulting set at $\alpha = 2/3$ has positive harmonic measure from every interior pole. The construction is exact. The note makes no claim of priority and records the interpretive boundary explicitly.

1. STATEMENT AND SCOPE

For $0 < \alpha < 1$, put

$$C_\alpha = \{X + iY : 0 < X < |Y|^\alpha\}.$$

This set has upper and lower connected components C_α^+ and C_α^- , distinguished by the sign of Y .

The source definition begins with angular boundary points. A finite point $z_0 \in \partial D$ is angular when some $\varepsilon > 0$ has the property that every component U of $D \cap B(z_0, \varepsilon)$ with $z_0 \in \partial U$ is contained in an angle of aperture less than π whose vertex is z_0 . Problem 7.15 then replaces the angles by translates of C_α . The sentence identifies the new comparison set but does not explicitly say what becomes of the vertex modifier.

We use the following finite interpretation class. It is deliberately wider than bare substitution, but it does not permit unrelated extra conditions.

Definition 1.1 (source-vocabulary readings). A reading in $\mathcal{R}_\alpha$ is obtained from the angular definition as follows.

- (1) Retain the tested point, the local radius, the local-component quantifier, the condition $z_0 \in \partial U$, and containment.
- (2) Replace the single angle comparator by a translate of C_α , C_α^+ , or C_α^- .
- (3) Delete the angle vertex condition, retain it as equality with the cusp tip, or weaken it canonically to boundary contact.
- (4) Allow equivalent definitions at the same value of α , including any nonempty union of the canonical closure cells in equation (4.1) below.
- (5) Allow the translating witness to be componentwise, common at one tested point, or common on the positive-measure witness set.

No new atomic predicate on $z_0 - w$ is allowed, and the unique comparator occurrence may not be copied, specialized at another parameter, and conjoined back into the generic definition.

Call a reading *endpoint-consistent* if it does not contradict the printed assertion that the zero-measure conclusion is false for a disk at $\alpha = 1/2$.

Theorem 1.2. *There is a bounded Jordan domain $D \subset \mathbb{C}$ such that, for every endpoint-consistent reading in $\mathcal{R}_{2/3}$, the corresponding boundary set has strictly positive harmonic measure from every pole $q \in D$. In particular, under the source-vocabulary convention of definition 1.1, the universal zero-harmonic-measure property in Problem 7.15 fails at $\alpha = 2/3$.*

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Remark 1.3. The theorem is conditional on the stated interpretation class. The source does not explicitly name the contact point of the translated cusp. A rule that adds an unprinted proper-sublocus condition is outside $\mathcal{R}_\alpha$ and can change the answer. Thus theorem 1.2 should not be quoted without its source-vocabulary qualification.

2. THE EXPLICIT JORDAN DOMAIN

Fix $\alpha = 2/3$, and write $C = C_{2/3}$. Define

$$\begin{aligned} U_+ &= \{x + iy : 1 < y < 3, 0 < x < y^{2/3}\}, \\ U_- &= \{x + iy : -3 < y < -1, 0 < x < (-y)^{2/3}\}, \\ R &= \{x + iy : 1/4 < x < 1/2, -2 < y < 2\}, \\ D &= U_+ \cup R \cup U_-. \end{aligned} \tag{2.1}$$

The point $p = 3/8$ belongs to D .

Proposition 2.1. *The set D is a bounded Jordan domain. Its boundary has positive logarithmic capacity.*

Proof. The three sets in equation (2.1) are open and connected, and

$$U_+ \cap R = \{1 < y < 2, 1/4 < x < 1/2\}, \quad U_- \cap R = \{-2 < y < -1, 1/4 < x < 1/2\}.$$

Hence their union is a domain. Its horizontal sections are

$$D_y = \begin{cases} (0, |y|^{2/3}), & 1 < |y| < 3, \\ (1/4, 1/2), & |y| \leq 1, \\ \emptyset, & |y| \geq 3. \end{cases} \tag{2.2}$$

For $1 < |y| < 2$, the rectangular section is already contained in the cusp section. Formula equation (2.2) exhibits the boundary as a simple closed chain consisting of four vertical lobe edges, two horizontal lobe caps, two curved graphs $x = |y|^{2/3}$, and the four sides joining the central bridge to the lobes. Nonconsecutive pieces are disjoint because the curved pieces have $x \geq 1$, the bridge sides have $x = 1/4$ or $1/2$, and the lobe sides have $x = 0$, with disjoint ordinate ranges where needed. Thus D is the bounded complementary component of a simple closed curve.

The segment $[i, 3i]$ lies in ∂D . A line segment of length L has logarithmic capacity $L/4$; consequently $\text{cap}(\partial D) \geq 1/2$. $\square$

3. FOUR POSITIVE-MEASURE CUSP ARCS

For $4/3 < u < 7/5$, set

$$\begin{aligned} E_{v,+} &= \{iu^3\}, & E_{c,+} &= \{u^2 + iu^3\}, \\ E_{v,-} &= \{-iu^3\}, & E_{c,-} &= \{u^2 - iu^3\}. \end{aligned} \tag{3.1}$$

These are open subarcs of ∂D . The subscripts v and c refer to the vertical and curved boundary strata of $\bar{C}$.

Lemma 3.1 (uniform local equality). *For every z_0 in any of the four arcs in equation (3.1),*

$$D \cap B(z_0, 1/8) = C \cap B(z_0, 1/8). \tag{3.2}$$

Proof. The curved points lie on $x = |y|^{2/3}$, since $(u^2)^3 = (u^3)^2$. The vertical points lie on $x = 0$, with nonzero imaginary part. Moreover,

$$\frac{64}{27} < |\text{Im } z_0| < \frac{343}{125}.$$

If $|z - z_0| < 1/8$, then $2 < |\text{Im } z| < 3$, with the same sign as $\text{Im } z_0$. On these two strips the bridge is absent and the appropriate lobe of D is exactly the corresponding component of C . This proves equation (3.2). $\square$

Proposition 3.2. *Each of the four arcs in equation (3.1) has strictly positive harmonic measure from every pole $q \in D$.*

Proof. Let $\phi : D \rightarrow \mathbb{D}$ be a Riemann map. By the Carathéodory boundary theorem, ϕ extends to a homeomorphism of the closures. Every arc E in equation (3.1) is nonempty and open in the Jordan curve ∂D , so $J = \phi(E)$ is a nonempty open circular arc. Conformal invariance and the disk Poisson kernel give

$$\omega_D^q(E) = \frac{1}{2\pi} \int_J \frac{1 - |\phi(q)|^2}{|\zeta - \phi(q)|^2} |d\zeta| > 0.$$

The integrand is positive and J has positive arclength. $\square$

4. NORMAL FORMS FOR THE REPLACEMENT CLAUSE

For a translate $w + C_\alpha$, write $z_0 - w = X + iY$. The closure of C_α has the seven canonical cells

$$\begin{array}{l|l} O & X = 0, Y = 0 \\ V_+ & X = 0, Y > 0 \\ V_- & X = 0, Y < 0 \\ \Gamma_+ & X = Y^\alpha, Y > 0 \\ \Gamma_- & X = (-Y)^\alpha, Y < 0 \\ I_+ & 0 < X < Y^\alpha, Y > 0 \\ I_- & 0 < X < (-Y)^\alpha, Y < 0. \end{array} \quad (4.1)$$

The full shape is compatible with all seven cells. The upper branch is compatible with O, V_+, Γ_+, I_+ , and the lower branch with O, V_-, Γ_-, I_- .

Lemma 4.1 (finite normal-form count). *Definition 1.1 has 157 nonempty compatible shape-selector pairs. With componentwise, pointwise-common, and witness-set-common translate scope, it has 471 scope-labelled normal forms.*

Proof. For the full shape, every nonempty union of the seven cells is allowed, giving $2^7 - 1 = 127$ selectors. Each branch has four compatible cells and therefore $2^4 - 1 = 15$ selectors. Thus there are $127 + 15 + 15 = 157$ shape-selector pairs. The three translate scopes multiply this number by three. $\square$

The endpoint assertion at $\alpha = 1/2$ restricts which selectors can represent the printed definition. Interior contact admits every point of a disk after a sufficiently distant translate. Curved contact admits a nondegenerate disk arc by the standard tangent-parabola computation. In contrast, origin contact makes each connected local component lie in one quadrant and hence makes the point angular; such points are harmonic-null. Vertical-only contact on a disk occurs only at an extremal point and is also harmonic-null. Consequently an endpoint-consistent selector contains at least one of

$$I_+, I_-, \Gamma_+, \Gamma_-. \quad (4.2)$$

5. ONE DOMAIN HANDLES EVERY SURVIVING READING

The arcs in equation (3.1), together with $w = 0$, handle the four nonorigin boundary cells:

$$V_+, V_-, \Gamma_+, \Gamma_-.$$

By lemma 3.1, the same local radius $1/8$ and the same translate work at every point of each respective arc.

For the two interior cells, put

$$w_+ = -1 - 100i, \quad w_- = -1 + 100i. \quad (5.1)$$

For $z = x + iy \in \overline{D}$, relative to either translate one has

$$1 \leq X = x + 1 < 4, \quad |Y| \geq 97 > 8 = 4^{3/2}.$$

Hence $X < |Y|^{2/3}$. The sign of Y is positive relative to w_+ and negative relative to w_- , so

$$\overline{D} \subset w_+ + I_+, \quad \overline{D} \subset w_- + I_-. \quad (5.2)$$

Thus a single translate handles every boundary point whenever the selector contains the corresponding interior cell.

Proof of theorem 1.2. Let an endpoint-consistent reading in $\mathcal{R}_{2/3}$ be given. By equation (4.2), its selector contains an interior or curved cell. If it contains Γ_+ or Γ_- , choose the corresponding curved arc in equation (3.1); lemma 3.1 and proposition 3.2 provide local containment and positive harmonic measure. If it contains I_+ or I_- , use equation (5.2); in this case all of ∂D is admitted. The argument uses a common translate, so it also satisfies every weaker componentwise or pointwise scope. Therefore every endpoint-consistent normal form produces a boundary set of positive harmonic measure. $\square$

6. WHY UNRESTRICTED ENRICHMENT IS DIFFERENT

The qualification in remark 1.3 is substantive. Consider

$$Q(X, Y) : \quad 0 < |Y| < 1, \quad X^2 = |Y|. \quad (6.1)$$

The condition $0 < |Y|$ can be formed after selecting nonorigin cells, but the cutoff $|Y| < 1$ is not supplied by the source. The equation $X^2 = |Y|$ is the curved-boundary equation only at $\alpha = 1/2$; at $\alpha = 2/3$ the corresponding equation is $X^3 = |Y|^2$. Conjoining both requires copying the unique comparator occurrence and specializing the copy at a second parameter. Hence Q is excluded by definition 1.1.

This exclusion cannot be replaced by the vague assertion that Q is unnatural. If arbitrary pragmatic additions are allowed, Q is a genuine endpoint-compatible rule that rejects the four frozen arcs, since their heights satisfy

$$\frac{64}{27} < |Y| < \frac{343}{125}$$

and therefore exceed one. The theorem concerns the finite occurrence-preserving, same-parameter source vocabulary, not unrestricted enrichment.

7. VERIFICATION AND STATUS

Two independent exact checks were performed. The geometric check verifies the strict parameter range, connected overlaps, horizontal sections, Jordan boundary, four arc types, local radius, curved-boundary identities, and the distant-translate inequalities using integer and rational arithmetic. The source-form check regenerates all 157 compatible selectors and all 471 scope-labelled normal forms, classifies the endpoint survivors, and checks that every surviving cell has a witness in equations (3.1) and (5.1).

The mathematical conclusion and the source-vocabulary qualification passed independent review. No claim is made that the interpretation is forced by an explicit contact convention absent from the printed sentence. No claim of novelty or priority is made. The value of the note is the exact domain, the robust finite interpretation audit, and the explicit statement of the remaining semantic boundary.

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